

GRUT Working Theory: Certified Structure, Conditional Emergence, and Open Primitives

A working-theory / program-status synthesis of the GRUT research program

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Abstract

This release is a working-theory / program-status synthesis of the GRUT research program. GRUT is currently a theory-construction / conditional-emergence research program with a certified dependency map over explicitly supplied physical layers; it is not currently a fundamental theory. The release states, with citations into the frozen repository record: the small set of structural relations derived from the program's core within declared model classes; the larger set of emergence results that hold only conditionally, after an explicitly supplied parent, sector, environment, preparation or lift is admitted; the primitives that remain supplied (among them the physical quantum lift, $\hbar$, outcome selection, and the gravitational branch); the routes that failed or are blocked (among them $\hbar$ emergence, Born-weight derivation at the tested model class, gravity forcing, the Π_0 cosmological route, the strict pointwise arrow, and the reversal hypothesis); and the frontiers that remain unresolved rather than failed (the physical origin of noise, and endogenous access). Every result carries its recorded model-class scope, and nothing in this release upgrades any recorded terminal.

Mandatory public statements: zero confirmed novel quantitative predictions · no external human peer review.

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Release provenance

Field	Value
Repository / branch	ryangrvr/grut-rai · master-w25bu9
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Release / packaging commit	the metadata-only descendant of 404f2cd recorded in CURRENT_STATE.json and cited in the release notes
Version lineage	next version on concept DOI 10.5281/zenodo.19803663; prior paper version DOI 10.5281/zenodo.22983638 (snapshot 6abbf31, 2026-09-26)
Authorization	Owner rulings SYN1-01 (Issue #2 comment 5916545551) and SYN1-02 (comment 5918358573)
Deposit status	Publication authorized; publication is not claimed until the returned DOI/version is recorded in the canonical state

This document renders three repository files with no content changes; in each part the file’s title line is replaced by the part heading. The repository snapshot at the release commit remains the deeper technical source; the supporting provenance documents (GRUT_WORKING_THEORY_SYNTHESIS_01.md, GRUT_RECORD_RECONCILIATION_INDEX_01.md, GRUT_SUCCESSOR_STATUS_01.md, SYN1_READINESS_VERDICT_01.md, SYN1_OWNER_RULING_01.md) are part of the release package in their Markdown form.

Part I — Working Theory Brief

(renders GRUT_WORKING_THEORY_PUBLIC_BRIEF_01.md)

A conservative public summary of where the GRUT research program stands. It is based only on results the program’s owner has accepted. The canonical record is GRUT_WORKING_THEORY_DEPOSIT_01.md; every statement below traces there.

Status of this brief: publication authorized (owner ruling SYN1-02, Issue #2 comment 5918358573); deposit pending. Publication is not claimed until the returned DOI/version is recorded in the canonical state.

What GRUT is, today

GRUT (historically, the Grand Responsive Universe Theory) set out to find a complete physical theory. After a long sequence of pre-registered, adversarially checked tests, its honest current form is narrower:

GRUT is a theory-construction research program. Its main product is a certified map of which assumptions produce which effective physical structures, where those derivations work only conditionally, and where they stop. It is not a fundamental theory.

The program describes its current architecture as a small set of **supplied layers**, each an input that the program does not derive. **Derivations run only downward** from these layers to effective laws and observables.

The supplied layers:

- a static substrate;
- a time-evolution law (the “generator”);
- how observations access the system;
- the state and preparation;
- an environment;
- the physical quantum realization (the “lift”);
- Planck’s constant $\hbar$;
- a rule for measurement outcomes;
- the gravitational sector.

This layering is bookkeeping, not a claim about how nature is built. It records where the current derivations stop.

What is derived

A small set of results follows from the program’s core assumptions alone. Each holds within a specific declared model class.

Core-derived:

- A spectral gap is necessary for the memory property studied.
- Passivity is necessary for the positivity property studied.
- Locality is necessary for the recovered geometric property in the tested class, but is not needed for the memory property.
- Linearity is what makes the exact reduction to linear spectral methods available. This one is definitional.
- The time-ordering found is a relabeling of the assumed time-evolution law, not an independent derivation of time.
- Two certified **non-implications**:
 - locality is not needed for the memory property;
 - the program’s earned static structure does not select a unique time-evolution law.

Further non-implications hold only in other declared model classes. One example is “emergent dissipation does not imply a strict ordering in time”.

What is conditional

These are genuine mathematical results, but each holds **only after a supplied layer is admitted**. None of them is “derived from nothing.”

- **Sector-conditioned laws.** Given a fixed microscopic law and a supplied conserved sector, different sectors lead to inequivalent large-scale effective laws. This was shown in a free-fermion model.
- **Emergent Markov behaviour.** Given a conservative microscopic model plus a controlled weak-coupling limit, a Markov (memoryless) law emerges. It is a *different* one from the law the program originally assumed.
- **Noise versus uncertainty.** In a declared nonlinear stochastic model, ongoing random forcing can be told apart observationally from uncertainty confined to the initial condition.
 - In the linear-Gaussian class the two remain equivalent.
 - The only exact deterministic re-description on record encodes the whole noise history in an enlarged hidden state. That is a mathematical representation, not a physical origin.
 - No declared physical environment reproduces it.

- **An integrated arrow of time.** In a declared infinite conservative chain with a declared initial temperature difference:
 - the system returns to equilibrium;
 - two declared bookkeeping measures each keep a net forward direction, even though microscopic reversals recur arbitrarily late:
 - * the integrated bath self-energy transfer, which carries an interaction-energy offset and is not a unique heat current;
 - * a reduced-state entropy-reference measure;
 - for the declared temperature pairs whose measure starts forward, cumulative forward progress is never erased.

This arrow is integrated, ensemble-level and preparation-relative. It is **not** a restored pointwise monotonic law, and it is not a derivation of a fundamental thermodynamic arrow.
- **Influence geometry.** Given $\hbar$ and a Gaussian model class, the realizable influence data form a specific convex “cone”.
- **Gravity sector.** Some gravitational results follow once the gravitational branch is supplied, but not before. The branch comprises the coupling class, spin-2, universal reach and a universal causal cone.
- **Cosmological kernel transport.** This is computable from admitted inputs.

What failed, or is blocked

These are the program’s recorded terminal outcomes, stated without softening:

- **The attempt to derive $\hbar$ failed.** $\hbar$ is an irreducible input: the program located where $\hbar$ enters its structure, but not its value. This terminal rests on multiply recorded verdicts; no standalone computation file survives.
- **Measurement outcomes and Born-rule weights were not derived** by the model class tested. This is not a disproof of the Born rule.
- **Gravity is underdetermined.** The program’s core does not select the gravitational coupling, spin-2, universal reach or a universal causal cone.
- **The proposed Π_0 cosmological route is blocked** at current scope. Π_0 is a proposed trace-channel response quantity. No calculable path to its prediction exists in the present record.
- **The physical quantum realization is not uniquely selected** by anything the program has earned. The program treats it as an irreducible supplied input, and it will not invent a new selector to rescue uniqueness.
- **The time-evolution law itself is not derived.** A conservative parent model yields a different law.
- **A strict, pointwise arrow of time is falsified** in the conservative model class. Only the integrated arrow above survives.
- **A hypothesized “reversal” structure** was not supported, meaning earned properties feeding back to become new ingredients.

What is unresolved (not failed)

- **The physical origin of noise.** Within the declared nonlinear model class, ongoing noise is distinguishable from initial uncertainty; in the linear-Gaussian class they remain equivalent. Whether a physical deterministic environment could produce it remains open: no declared candidate exists yet.
- **Whether observational access can be derived from the classical substrate.** The

question cannot yet be posed at the program’s classical core level. It has not been answered negatively.

What remains supplied

These enter as inputs, not results:

- the substrate;
- the time-evolution law and its direction;
- how access and readout are defined;
- the conserved sector and the initial preparation;
- the environment and coarse-graining;
- the physical quantum realization and its structures (statistics, complex structure, and so on);
- $\hbar$;
- the measurement-outcome rule;
- the gravitational sector;
- the cosmological transport inputs.

What is experimentally predictive today

The program currently has zero confirmed novel quantitative predictions, and its results have not had external peer review.

- The conditional calculations above are internal mathematical results, not empirical predictions.
- The one cosmological route proposed as a percent-level prediction (Π_0) is blocked at current scope.
- An earlier set of candidate observational channels (the “v4” set) was closed with its channels still open, not confirmed.

Where this leaves the program

GRUT does not currently claim a final theory. It has a defensible working account:

- a small core of earned structural relations;
- a larger set of conditional emergence theorems built on explicitly supplied physical layers;
- a clear record of where the derivations stop.

An architecturally central gap is the step from a classical substrate to the state/observation structure that quantum physics requires. That gap is recorded, not closed.

For sources and exact wording, see `GRUT_WORKING_THEORY_DEPOSIT_01.md`, `GRUT_WORKING_THEORY_SYNTHESIS_01.md` and the owner rulings they cite.

Part II — Canonical Working-Theory Deposit

(renders `GRUT_WORKING_THEORY_DEPOSIT_01.md`, SYN-1 D-1)

STATUS: DEPOSIT. This freezes the current theory boundary.

Rules:

- **No new analysis and no synthesis beyond SYN-0.** Every entry is a pointer into accepted records.

- **Authority:** SYNO_OWNER_RULING_01.md (Issue #2 comment 5915940892).
- **Governing synthesis:** GRUT_WORKING_THEORY_SYNTHESIS_01.md at 91eaefd.
- **Companion files:**
 - GRUT_RECORD_RECONCILIATION_INDEX_01.md (stale wording → the governing record);
 - GRUT_SUCCESSOR_STATUS_01.md (live successor status).

Date: 2026-09-30 · **Branch:** master-w25bu9.

A. Current identity

GRUT is presently a theory-construction program whose accepted product is a certified dependency map of effective structure over an explicitly supplied, multi-layer foundation. It is not a fundamental theory, and it is not only an EFT. (SYNO_OWNER_RULING_01.md §2, verbatim)

Fence: the nine-layer architecture (§E) is the **minimal bookkeeping architecture supported by the current record**, not a claim that nature is fundamentally partitioned into exactly nine layers (SYNO_OWNER_RULING_01.md §2).

B. Supplied ledger (by reference)

See SYN-0 §1, **Ledger A (A-1 ... A-16)**, where each entry is cited to the ruling that leaves it supplied.

Summary:

- the static substrate and the local net;
- the temporal generator, its orientation, and the declared nonlinear drift;
- the access seed, readout and access sets;
- the conserved sector, statistics, initial preparation and partition;
- the physical environment (noise origin) and coarse-graining;
- the physical quantum lift and its prices;
- physical $\hbar$;
- the outcome-selection / Born rule;
- the gravitational branch (coupling class, spin-2, universal reach, universal causal cone);
- the cosmological transport inputs.

Price rule: “Lorentz/gauge, symplectic, complex structure, statistics, noise law and physical $\hbar$ distinguish classes only after they are supplied” (S5_OWNER_RULING_01.md:85–86).

C. Earned ledger

C-1 Core-derived (accepted; SYNO_OWNER_RULING_01.md §3)

Item	Scope / grade	Source
gap → memory	LOAD-BEARING	LO_1A_OWNER_RULING_01.md
passivity → positivity	LOAD-BEARING	LO_1A_OWNER_RULING_01.md
locality → geometry	NECESSITY-CERTIFIED	LO_1B_OWNER_RULING_01.md
linearity → exact reduction	NECESSITY-CERTIFIED (definitional)	LO_1C_OWNER_RULING_01.md
locality $\nRightarrow$ memory	—	LO_1B_OWNER_RULING_01.md
the O-5 clock/order	a relabeling of the supplied generator	LO_1F_OWNER_RULING_01.md

Item	Scope / grade	Source
end-site observability	identity-grade (TRIVIAL/IDENTITY)	LO_ACCESS_BRIDGE_OWNER_RULING_02.md
earned static structure $\Rightarrow$ unique temporal generator	—	S5_OWNER_RULING_01.md:81

All of these hold on the declared C1-a class, at recorded scope.

Class-scoped floor results (not core-alone):

- cycle affinity $\rightarrow \neg P^{\text{resp}}_{\text{CM}}$ (L0-1d ring);
- linearity $\Rightarrow$ memory (L0-1c convex-quartic class). This is the accepted content of the “nonlinear relaxation class”: a **deletion** result, transient-limited, and not a selected class;
- emergent dissipation $\Rightarrow$ strict ordering (O-6 conservative parent);
- the remaining floor non-implications (LO_1_FLOOR_DEPOSIT_01.md:95–102), excluding locality $\Rightarrow$ memory, which is core.

C-2 Conditionally emergent: derived given explicitly supplied upstream structure

Result	Terminal	Supplied upstream	Record
Sector-conditioned law formation	SF-1 = FORMATION-OF-LAW-CLASS	conserved sector, free-fermion parent, statistics	SF1_OWNER_RULING_02.md
Underdamped Markov emergence	S5-1 = MARKOV-LIMIT-OTHER-CLASS	conservative parent + admitted L-vH	S5_OWNER_RULING_03.md
Nonlinear stochastic discriminator	S2-1 = FULL-DISCRIMINATOR-CONFIRMED	L0-1c drift + declared additive noise	S2_OWNER_RULING_03.md
Return to equilibrium; integrated arrow	S6-1 = NET-ARROW-CONFIRMED (NO-ERASURE-ON-OPEN-MEMBERS)	conservative parent + declared preparation; J = bath self-energy flux	S6_OWNER_RULING_02.md
Influence-cone geometry	P-2 CONE-CONFIRMED (Gaussian class)	$\hbar$ / quantum class	P2_S1_OWNER_RULING_01.md
Gravity-sector results (U-1, RS-1, TT-1, ω^7 occupancy)	conditional	the supplied quadruple	GR2_SYNTHESIS_OWNER_RULING_01.md; GR2A_OWNER_RULING_01.md
FRW kernel transport	computable	admitted inputs	KERNEL_TRANSPORT_OWNER_RULING_01.md

Also recorded, but not in the owner’s §3 list: branch-class selection, PROVISIONAL (in class). See GRUT_RECORD_RECONCILIATION_INDEX_01.md, item 10.

These are not “derived from GRUT from nothing.”

D. Hard negative ledger (exact terminals)

Failed / blocked / falsified

Terminal (exact)	Record
$\hbar$ emergence: FAILED , $\hbar$ is an irreducible input (“located, not derived”; P-2: “Located, not generated”)	program/GRUT_REALITY_CHECK_01.md:44; GRUT_WORKING_THEORY_01.md:311; P2_S1_OWNER_RULING_01.md:37-43 GRUT_PROGRAM_CLOSURE_01.md:38-43
EXPERIMENT_P_FRONTIER = CLOSED_AT_TESTED_MODEL_CLASS (“not a universal disproof of the Born rule”); outcome selection / Born weights not derived	
GRAVITY_UNDERDETERMINED	GR2_SYNTHESIS_OWNER_RULING_01.md:13-14
$\Pi 0-0 = \text{FRONTIER-BLOCKED}$	PIO_TRACE_CHANNEL_OWNER_RULING_01.md:11-12
L0 ACCESS BRIDGE = NONUNIQUE-LIFT;	L0_ACCESS_BRIDGE_OWNER_RULING_02.md:10;
L0 LIFT SELECTION = IRREDUCIBLE/SUPPLIED	L0_LIFT_SELECTION_OWNER_RULING_02.md:9
REV-0 = NOT SUPPORTED at recorded scope	L0_STRUCTURAL_DIAGNOSTIC_REVERSAL_OWNER_RULING_01.md:9
O-6 = FALSIFIED (strict pointwise arrow)	L0_1G_OWNER_RULING_02.md:9-10
O-2 = FALSIFIED; O-7 = FALSIFIED	L0_1H_OWNER_RULING_03.md:8; L0_1_FLOOR_07_OWNER_RULING_01.md:10
S5-0 = GENERATOR-IRREDUCIBLE/SUPPLIED ; the conservative parent does not derive the Level-0 G-D generator	S5_OWNER_RULING_01.md:9; S5_OWNER_RULING_03.md:117-120
SFG-0 = CLASS-SPLIT (Tier E: no formation variable; “NO CURRENT CANDIDATE”)	SFG0_OWNER_RULING_01.md:60,73
S3-0 = FORMULABLE-ONLY-WITH-CHANGE (the crossed cell is not in the theory at fixed invariants)	S3_OWNER_RULING_01.md:9

The full list is in SYN-0 §4.

Unresolved (not failed)

Terminal (exact)	Record
Physical origin of the noise: S2-HB = UNRESTRICTED-REALIZATION-ONLY , sub-label NO-PHYSICAL-BATH-CANDIDATE	S2_HB_OWNER_RULING_02.md:9,47
Endogenous access: EA-0 = UNFORMULABLE at the earned Level-0 scope (“not negatively answered”)	EA0_OWNER_RULING_02.md:11-21

Prediction status

Zero confirmed novel quantitative predictions; no external human peer review (GRUT_PROGRAM_STATE_SYNTHESIS_01.md:164-165). v4 is DEPOSITED_CLOSED, with its channels “STILL OPEN” (V4_DEPOSIT_01.md).

E. Minimal architecture (accepted; SYN0_OWNER_RULING_01.md §5)

substrate + generator + access + state/sector/preparation + environment + physical lift → effective observables/laws

$\hbar$, the outcome rule and the gravitational branch are carried as separately supplied structures where required.

- **The nine supplied layers** are listed in SYNO_OWNER_RULING_01.md §2. They are **mutually non-selecting** on the record. The only cross-layer relations are possibility without selection (NONUNIQUE-LIFT) and conditional supply (statistics $\rightarrow$ sector, SFG-0 Tier C). Physical $\hbar$ is also one of the lift's prices, so those two layers overlap.
- **Arrow grades** are DERIVED / CONDITIONAL / SUPPLIED / BLOCKED / FALSIFIED-NON-IMPLICATION: 14 graded arrows and 18 graded non-arrows, in SYN-0 §5.2.
- **Direction:** all derivations run downward, and no property $\rightarrow$ ingredient edge exists (REV-0, recorded scope).
- **No reversal/self-duality diagram.**
- **Architecturally central seam:** classical local substrate ? state/access algebra (EA0_OWNER_RULING_02.md:110–112). **It is not an open campaign.** L0 LIFT SELECTION = IRREDUCIBLE/SUPPLIED stands. The preserved faithful-representation audit could only constrain, not select.

F. Claim fence

GRUT currently claims:

- a small core of certified ingredient $\rightarrow$ property relations and non-implications, in declared classes;
- a set of conditional emergence theorems, each valid given its explicitly supplied upstream structure;
- a hard, cited map of what failed, what is blocked, and what remains unresolved;
- the minimal bookkeeping architecture above.

GRUT does not currently claim:

- to be a fundamental theory, or a theory of everything;
- to derive $\hbar$, the physical quantum lift, the Born rule or outcome selection, or gravity (coupling, spin-2, universal reach, causal cone);
- a Π_0 -based cosmological prediction;
- that noise is primitive, or that it is not;
- a fundamental thermodynamic arrow, or broken microscopic reversibility. The S6 arrow is integrated, ensemble-level and preparation-relative;
- that J is a unique heat current (it is the bath self-energy flux);
- that any earned property constructs a supplied ingredient;
- that nature consists of exactly nine layers;
- any confirmed novel quantitative prediction;
- that the temporal generator or the L0-1c drift is derived;
- that coarse-graining is dynamically derived;
- that O-6 is repaired;
- that GRUT has a formation variable;
- a disproof of the Born rule;
- that reality is nonlocal.

Part III — Publication Manifest

(renders GRUT_PUBLICATION_MANIFEST_01.md)

Authority: SYN1_OWNER_RULING_02.md §8 (Issue #2 comment 5918358573). **This manifest contains no new scientific analysis.**

Field	Value
Public title	GRUT Working Theory: Certified Structure, Conditional Emergence, and Open Primitives
Release identity	Working-theory / program-status synthesis (not a ToE claim, not a prediction paper, not peer-reviewed)
Manifest date	2026-09-30
Publication date	Assigned at deposit; recorded with the returned DOI/version
Repository	<code>ryangrvt/grut-rai</code>
Branch	<code>master-w25bu9</code>
Frozen content boundary (exact commit)	<code>404f2cd</code> — the publication-ready state (5/5 editorial cross-check)
Release / packaging commit	The descendant of <code>404f2cd</code> that adds only this manifest, the publication rulings and state bookkeeping. <code>gitlog-1--GRUT_PUBLICATION_MANIFEST_01.md</code> identifies it; the release notes must cite it explicitly.
Version lineage	The next version in the existing GRUT public record lineage, concept DOI 10.5281/zenodo.19803663 . Prior paper version: DOI 10.5281/zenodo.22983638 (snapshot <code>6abbf31</code> , 2026-09-26, with the dated update of 2026-09-27). Software concept lineage: DOI 10.5281/zenodo.18993689.

Included documents

Primary public documents:

1. `GRUT_WORKING_THEORY_DEPOSIT_01.md` — the canonical working-theory deposit
2. `GRUT_WORKING_THEORY_PUBLIC_BRIEF_01.md` — the public-facing brief

Supporting provenance / navigation:

3. `GRUT_WORKING_THEORY_SYNTHESIS_01.md` — the accepted SYN-0 synthesis (five ledgers; architecture)
4. `GRUT_RECORD_RECONCILIATION_INDEX_01.md` — canonical pointers for stale/conflicting historical wording
5. `GRUT_SUCCESSOR_STATUS_01.md` — the live successor-status index
6. `SYN1_READINESS_VERDICT_01.md` — the readiness verdict, with the editorial cross-check appendix
7. `SYN1_OWNER_RULING_01.md` — the owner ruling fixing the A-1 ... A-4 statuses

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